

Use this document to detail specifications/requirements for an item and/or ancillary services. ⓘ

TITLE: Multi-channel 12 GHz microwave synthesizer ⓘ

Requesting Lab/Division/Group: PML/Time and Frequency, Atomic Devices and Instrumentation, 688.90

General Statement of Need ⓘ

For experiments on chip-scale atomic clocks based on atomic Cs, we require a low-phase noise microwave synthesizer with capabilities to produce 5 independent signals up to 12 GHz in frequency. For phase stability between the outputs, we need the signals to be generated from a single instrument capable of producing 3 independent signal outputs in the 10 MHz to 12 GHz range and 2 additional independent outputs in the 10 MHz to 6 GHz range. The synthesizer must be able to be referenced to external 10 or 100 MHz frequency signals and have frequency resolution of at least 0.001 Hz. The instrument will be controlled via a computer and doesn't necessarily need a front panel interface, but it will need a graphical user interface for computer control. The instrument needs to have the ability to perform digital frequency and amplitude sweeps using a list or similar control structure.

Items and/or Ancillary Services ⓘ

Item 1: Microwave synthesizer

Quantity: 1

Specifications: See detailed specification in attached "Detailed specification document".

Separately Priced Options and Services:

Item 2: [List minimum specifications.]

Quantity: [Enter quantity here.]

Specifications: [Enter quantity here.]

[List any other item(s) or service requirements here using the same format above.]

Service Requirements / Item Specifications

Section ⓘ	Requirements/Specifications ⓘ
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain requirement here] ☐ US citizen required on-site (foreign nationals require preregistration 30 days prior) ☐ Vendor rep on-site more than three days ☐ Other considerations: [List other considerations to ensure an accurate quote]
Shipping & Delivery ☒ N/A	[Enter shipping and delivery information by item here.] Shipping criteria: ☐ Partial delivery acceptable

Requirements / Specifications: Multi-channel 12 GHz microwave synthesizer

1

Version 1.0 (2019)

Double-click on ⓘ for further assistance.

☐ Direct delivery to building required [Explain need for requirement here] Building has: ☐ Loading dock ☐ Containerization preference: [Choose container] Other requirements: [List any other requirements here] Delivery date criteria: ☒ Delivery of goods, installation, and training shall be completed no later than: 12 weeks ARO ☐ Deliver no earlier than: [Insert date here] ☐ Other: [List other shipping, delivery, or specil requirements]	
Electronic Media ☐ N/A	GUI software for control of microwave synthesizer. ☒ Software ☐ Electronic manuals ☐ License Key ☐ Other: [List any other Electronic Media requirements here] [Enter email address for software delivery here]
Installation ☒ N/A	[Enter installation requirements by item here.] ☐ Vendor technician on site ☐ Rigging ☐ Uncrating / Unpackaging ☐ Removal of packaging material ☐ Equipment set up ☐ Start-up services ☐ Turn key installation ☐ Other [Enter details here.] Note: When installation is required, TPOC or COR shall be the coordination point.
Facility / Utility Considerations ☒ N/A	[List site specific utility considerations by item here.] ☐ OFPM Work Order Number(s) [List any Work Order Number(s) here] ☐ Power available Voltage: [Enter voltage here] (V) Max Current: [Enter max current draw here] (A) Phase: [Choose phase] Frequency: [Typically 60Hz] (Hz) NEMA Plug Type: [Enter plug type here] (i.e.: NEMA L5-20P) Environment: [Choose environment] Other power consideration: [List any other power considerations here] ☐ Utility connections: [List relevant available utilities, pressure(s), and flowrate(s)] ☐ Other considerations: [List other considerations to ensure an accurate quote]

Warranty ⓘ

☐ N/A

Base Manufacturer Warranty is acceptable

[If you require an Extended Manufacturer Warranty, describe the warranty requirements per item here.]

What is included?

☐ Labor ☐ Parts ☐ Travel ☐ Cost and liability for returns

☐ Other: [Enter other requirements here]

☐ Warranty length: [Enter warranty length here]

Period Length:[Ex. month, 6 month, year etc.]

Training

☒ N/A

[Training requirements (who, when, what, where) here. If on-site, is travel necessary?]

☐ On-site

☐ Off-site location: [Enter location(s) here]

☐ Number of trainees: [Enter number of trainees here]

Scope of training:

☐ Operations ☐ Maintenance ☐ Troubleshooting ☐ Safety

☐ Other: [Enter other scope requirements here]

☐ Training materials to be provided: [Describe requirement here]

☐ Other: [Enter other training requirements here]

Note: When training is required, TPOC or COR shall be the coordination point.

Inspection and Acceptance

☐ N/A

Will measure synthesizer output frequency and power

☒ Planned government acceptance testing (7 business days)

☐ Gov't acceptance expected to take longer than 7 business days: [Duration]

☒ Inspection and acceptance plan: Will measure synthesizer output frequency and power

☐ Other: [Enter other requirements here]

Maintenance ⓘ

☒ N/A

[Describe your maintenance requirements per item here]

Period Length: [Ex. month, 6 month, year etc.]

What is included?

☐ Labor ☐ Spare parts ☐ Consumables ☐ Travel

☐ Cost and liability for returns ☐ Software updates ☐ Manuals

☐ Service plan length: [Enter service plan length here]

☐ Other: [Enter other requirements here]

☐ Requirements: [Enter requirements here] (i.e.: onsite within 72 hours, etc.)

☐ Option Period 1 ☐ Option Period 2

☐ Option Period 3 ☐ Option Period 4

Government Furnished Material / Equipment

[List any government furnished material/equipment per item here]

☐ Samples for testing

☐ Government equipment which will leave gov't site (TPOC must coordinate with division property custodian)

Serial Number: [Enter Serial Number here]

☒ N/A

Description: [Enter description here]

NIST property number: [Enter NIST property number here]

☐ Other: [Enter other details here]

Travel

[Describe any travel requirements per item here]

☒ N/A

☐ Explain: [Explain here]

☐ Other: [Enter other requirements here]

Vendor: To the maximum extent possible, include as part of appropriate sections above (maintenance, repair, etc.). Otherwise, it will be a separate line item, reimbursable per the Federal Travel Regulation.

Provide suggestions to improve this form by clicking [here](#).